What might the deck be?

Year 6 Statistics and Probability · Lesson 3 · AC9M6P01 · Version S

NAMES _____ CLASS _____

THE PROBLEM

Every card in a deck is either a picture card or a numbered card.
If you draw one card without looking, the probability of a
PICTURE CARD is 1/4.
What might the deck be?

Part A · My first deck

Work on your own for the first five minutes.

PICTURE CARDS	EVEN CARDS	ODD CARDS	TOTAL	DOES IT WORK?

Part B · Test it

Draw a card, record it, then PUT IT BACK. Twenty times.

TICK IF IT WAS A PICTURE CARD

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

I drew a picture card _____ times out of 20.

WORKING SPACE — SKETCH YOUR DECK HERE

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Part C · More than one answer

Find other decks that also work. Fill in a row for each one.

PICTURE CARDS	EVEN CARDS	ODD CARDS	TOTAL	DOES IT WORK?

Check BOTH probabilities before you write "yes".

Part D · Show your thinking

SF **1. What is the smallest deck that works?**

SF **2. Draw a deck of 8 cards that works.**

Write how many of each kind you used.

CU **3. What do the totals of all the decks that work have in common?**

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Exit ticket

Hand this in at the end of the lesson.

A bag of counters has a 1/5 chance of drawing a RED counter.

1. Give two DIFFERENT bags that would work.

They must have different numbers of counters altogether.

BAG 1

BAG 2

2. Could the bag hold 8 counters? Explain your answer.

3. In one of your bags, what is the probability of drawing a counter that is NOT red?

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